

HackerRank Java Programs

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1. Question :

HackerRank | Prepare > Java > Introduction Welcome to Java!

Problem

Welcome to the world of Java! In this challenge, we practice printing to stdout.

The code stubs in your editor declare a `Solution` class and a main method. Complete the main method by copying the two lines of code below and pasting them inside the body of your main method.

```
System.out.println("Hello, World.");
System.out.println("Hello, Java.");
```

Input Format

There is no input for this challenge.

Output Format

You must print two lines of output:

1. Print `Hello, World.` on the first line.
2. Print `Hello, Java.` on the second line.

Sample Output

```
Hello, World.
Hello, Java.
```

Submissions

Leaderboard

Code :

```
public class Solution {

    public static void main(String[] args) {
        System.out.println("Hello, World.");
        System.out.println("Hello, Java.");
    }
}
```

Output :

Congratulations!
You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Your Output (stdout)

```
1 Hello, World.
2 Hello, Java.
```

Expected Output

```
1 Hello, World.
2 Hello, Java.
```

Download

2. Question :

Task

In this challenge, you must read 3 integers from stdin and then print them to stdout. Each integer must be printed on a new line. To make the problem a little easier, a portion of the code is provided for you in the editor below.

Input Format

There are 3 lines of input, and each line contains a single integer.

Sample Input

```
42
100
125
```

Sample Output

```
42
100
125
```

Code :

```
import java.util.*;

public class Solution {

    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        int a = scan.nextInt();
        int b = scan.nextInt();
        int c = scan.nextInt();

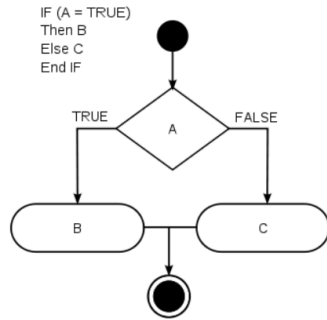
        System.out.println(a);
        System.out.println(b);
        System.out.println(c);
    }
}
```

Output :

Congratulations
You solved this challenge. Would you like to challenge your friends? [f](#) [t](#) [in](#) [Next Challenge](#)

✔ Test case 0	Compiler Message	
✔ Test case 1	Success	
✔ Test case 2	Input (stdin)	Download
	<pre>1 42 2 100 3 125</pre>	
	Expected Output	Download
	<pre>1 42 2 100 3 125</pre>	

3 . Question :



Source: Wikipedia

Task

Given an integer, n , perform the following conditional actions:

- If n is odd, print `Weird`
- If n is even and in the inclusive range of 2 to 5, print `Not Weird`
- If n is even and in the inclusive range of 6 to 20, print `Weird`
- If n is even and greater than 20, print `Not Weird`

Complete the stub code provided in your editor to print whether or not n is weird.

Input Format

A single line containing a positive integer, n .

Code :

```
1  import java.io.*;
2  import java.math.*;
3  import java.security.*;
4  import java.text.*;
5  import java.util.*;
6  import java.util.concurrent.*;
7  import java.util.regex.*;
8  import java.util.Scanner;
9
10 public class Solution {
11
12
13
14     private static final Scanner scanner = new Scanner(System.in);
15
16     public static void main(String[] args) {
17         int N = scanner.nextInt();
18         scanner.skip("(\\r\\n|\\[\\n\\r\\u2028\\u2029\\u0085])?");
19         if (N%2!=0){
20             System.out.println("Weird");
21         }else if (N%2==0 && N>1 && N<6){
22             System.out.println("Not Weird");
23         }else if (N%2==0 && N>5 && N<21){
24             System.out.println("Weird");
25         }else if (N%2==0 && N>20){
26             System.out.println("Not Weird");
27         }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0	Input (stdin)	Download
✔ Sample Test case 1	1 3	
	Your Output (stdout)	
	1 Weird	
	Expected Output	Download
	1 Weird	

4 . Question :

In this challenge, you must read an integer, a double, and a String from stdin, then print the values according to the instructions in the Output Format section below. To make the problem a little easier, a portion of the code is provided for you in the editor.

Note: We recommend completing [Java Stdin and Stdout I](#) before attempting this challenge.

Input Format

There are three lines of input:

1. The first line contains an integer.
2. The second line contains a double.
3. The third line contains a String.

Output Format

There are three lines of output:

1. On the first line, print `String`: followed by the unaltered String read from stdin.
2. On the second line, print `Double`: followed by the unaltered double read from stdin.
3. On the third line, print `Int`: followed by the unaltered integer read from stdin.

To make the problem easier, a portion of the code is already provided in the editor.

Code :

```
1  import java.util.Scanner;
2  import java.lang.String;
3
4  public class Solution {
5
6      public static void main(String[] args) {
7          Scanner scan = new Scanner(System.in);
8          int i = scan.nextInt();
9          double d = scan.nextDouble();
10         scan.nextLine();
11         String s = scan.nextLine();
12
13
14         System.out.println("String: " + s);
15         System.out.println("Double: " + d);
16         System.out.println("Int: " + i);
17     }
18 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Sample Test case 1

Input (stdin)

```
1 42
2 3.1415
3 Welcome to HackerRank's Java tutorials!
```

Download

Your Output (stdout)

```
1 String: Welcome to HackerRank's Java tutorials!
2 Double: 3.1415
3 Int: 42
```

Expected Output

```
1 String: Welcome to HackerRank's Java tutorials!
2 Double: 3.1415
```

Download

5 . Question :

Java's `System.out.printf` function can be used to print formatted output. The purpose of this exercise is to test your understanding of formatting output using `printf`.

To get you started, a portion of the solution is provided for you in the editor; you must format and print the input to complete the solution.

Input Format

Every line of input will contain a `String` followed by an integer.

Each `String` will have a maximum of 10 alphabetic characters, and each integer will be in the inclusive range from 0 to 999.

Output Format

In each line of output there should be two columns:

The first column contains the `String` and is left justified using exactly 15 characters.

The second column contains the integer, expressed in exactly 3 digits; if the original input has less than three digits, you must pad your output's leading digits with zeroes.

Code :

```
1 import java.util.Scanner;
2 import java.lang.String;
3
4 public class Solution {
5
6     public static void main(String[] args) {
7         Scanner sc=new Scanner(System.in);
8         System.out.println("=====");
9         for(int i=0;i<3;i++){
10             String s1=sc.next();
11             int x=sc.nextInt();
12             System.out.printf("%-15s%03d%n",s1,x);
13         }
14
15
16         System.out.println("=====");
17
18     }
19 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

Input (stdin)

```
1 java 100
2 cpp 65
3 python 50
```

Your Output (stdout)

```
1 =====
2 java          100
3 cpp           065
4 python        050
5 =====
```

6 . Question :

Task

Given an integer, N , print its first 10 multiples. Each multiple $N \times i$ (where $1 \leq i \leq 10$) should be printed on a new line in the form: $N \times i = \text{result}$.

Input Format

A single integer, N .

Constraints

- $2 \leq N \leq 20$

Output Format

Print 10 lines of output; each line i (where $1 \leq i \leq 10$) contains the *result* of $N \times i$ in the form:

$N \times i = \text{result}$.

Code :

```
1 import java.io.*;
2 import java.math.*;
3 import java.security.*;
4 import java.text.*;
5 import java.util.*;
6 import java.util.concurrent.*;
7 import java.util.regex.*;
8
9
10
11 public class Solution {
12     public static void main(String[] args) throws IOException {
13         BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
14
15         int N = Integer.parseInt(bufferedReader.readLine().trim());
16
17         for (int i=1;i<11;i++){
18             int mul=1;
19             mul=N*i;
20             System.out.println(N+" x "+i+" = "+mul);
21         }
22
23         bufferedReader.close();
24     }
25 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

1 2

Your Output (stdout)

```
1 2 x 1 = 2
2 2 x 2 = 4
3 2 x 3 = 6
4 2 x 4 = 8
5 2 x 5 = 10
6 2 x 6 = 12
7 2 x 7 = 14
8 2 x 8 = 16
9 2 x 9 = 18
10 2 x 10 = 20
```

7 . Question :

We use the integers a , b , and n to create the following series:

$$(a + 2^0 \cdot b), (a + 2^0 \cdot b + 2^1 \cdot b), \dots, (a + 2^0 \cdot b + 2^1 \cdot b + \dots + 2^{n-1} \cdot b)$$

You are given q queries in the form of a , b , and n . For each query, print the series corresponding to the given a , b , and n values as a single line of n space-separated integers.

Input Format

The first line contains an integer, q , denoting the number of queries.

Each line i of the q subsequent lines contains three space-separated integers describing the respective a_i , b_i , and n_i values for that query.

Constraints

- $0 \leq q \leq 500$
- $0 \leq a, b \leq 50$
- $1 \leq n \leq 15$

Output Format

For each query, print the corresponding series on a new line. Each series must be printed in order as a single line of n space-separated integers.

Code :

```
1 import java.util.*;
2 import java.io.*;
3
4 class Solution{
5     public static void main(String []argh){
6         Scanner in = new Scanner(System.in);
7         int t=in.nextInt();
8         for(int i=0;i<t;i++){
9             int a = in.nextInt();
10            int b = in.nextInt();
11            int n = in.nextInt();
12            int sum=a;
13            for (int j=0;j<n;j++){
14                sum+=(1 << j)*b;
15                System.out.print(sum + " ");
16            }
17            System.out.println();
18        }
19        in.close();
20    }
21 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Sample Test case 1

Input (stdin)

```
1 2
2 0 2 10
3 5 3 5
```

Your Output (stdout)

```
1 2 6 14 30 62 126 254 510 1022 2046
2 8 14 26 50 98
```

Expected Output

```
1 2 6 14 30 62 126 254 510 1022 2046
2 8 14 26 50 98
```

8. Question :

Java has 8 primitive data types; char, boolean, byte, short, int, long, float, and double. For this exercise, we'll work with the primitives used to hold integer values (byte, short, int, and long):

- A byte is an 8-bit signed integer.
- A short is a 16-bit signed integer.
- An int is a 32-bit signed integer.
- A long is a 64-bit signed integer.

Given an input integer, you must determine which primitive data types are capable of properly storing that input.

To get you started, a portion of the solution is provided for you in the editor.

Reference: <https://docs.oracle.com/javase/tutorial/java/nutsandbolts/datatypes.html>

Input Format

The first line contains an integer, T , denoting the number of test cases.

Each test case, T , is comprised of a single line with an integer, n , which can be arbitrarily large or small.

Code :

```
6  class Solution{
7      public static void main(String []argh)
8
9
10
11
12      Scanner sc = new Scanner(System.in);
13      int t=sc.nextInt();
14
15      for(int i=0;i<t;i++)
16      {
17
18          try
19          {
20              long x=sc.nextLong();
21              System.out.println(x+" can be fitted in:");
22              if(x>=Byte.MIN_VALUE && x<= Byte.MAX_VALUE)System.out.println("* byte");
23              if(x>=Short.MIN_VALUE && x<=Short.MAX_VALUE)System.out.println("* short");
24              if(x>=Integer.MIN_VALUE && x<=Integer.MAX_VALUE)System.out.println("* int");
25              if(x>=Long.MIN_VALUE && x<=Long.MAX_VALUE)System.out.println("* long");
26          }
27          catch(Exception e)
28          {
29              System.out.println(sc.next()+" can't be fitted anywhere.");
30          }
31      }
32  }
33 }
34 }
```


Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Your Output (stdout)

[illegible]

9. Question :

The challenge here is to read n lines of input until you reach EOF, then number and print all n lines of content.

Hint: Java's `Scanner.hasNext()` method is helpful for this problem.

Input Format

Read some unknown n lines of input from `stdin(System.in)` until you reach EOF; each line of input contains a non-empty String.

Output Format

For each line, print the line number, followed by a single space, and then the line content received as input.

Code :

```

1  import java.io.*;
2  import java.util.*;
3  import java.text.*;
4  import java.math.*;
5  import java.util.regex.*;
6  import java.util.Scanner;
7
8  public class Solution {
9
10     public static void main(String[] args) {
11         Scanner sc=new Scanner(System.in);
12         int n=1;
13         while (sc.hasNext()){
14             String l=sc.nextLine();
15             System.out.println(n + " " + l);
16             n++;
17         }
18         sc.close();
19     }
20 }

```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

```
1 Hello world
2 I am a file
3 Read me until end-of-file.
```

Your Output (stdout)

```
1 1 Hello world
2 2 I am a file
3 3 Read me until end-of-file.
```

10. Question :

You are given an integer n , you have to convert it into a string.

Please complete the partially completed code in the editor. If your code successfully converts n into a string s the code will print "Good job".

Otherwise it will print "Wrong answer".

n can range between -100 to 100 inclusive.

Sample Input 0

100

Code :

```
4 import java.util.*; ...
13 String s= Integer.toString(n);
14
15 ✓
16 if (n == Integer.parseInt(s)) {
17     System.out.println("Good job");
18 } else {
19     System.out.println("Wrong answer.");
20 }
21 } catch (DoNotTerminate.ExitTrappedException e) {
22     System.out.println("Unsuccessful Termination!!");
23 }
24 }
25 }
26
27 //The following class will prevent you from terminating the code using exit(0)!
28 class DoNotTerminate {
29
30     public static class ExitTrappedException extends SecurityException {
31
32         private static final long serialVersionUID = 1;
33     }
34
35     public static void forbidExit() {
36         final SecurityManager securityManager = new SecurityManager() {
37             @Override
38             public void checkPermission(Permission permission) {
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Input (stdin)	
1	100
Your Output (stdout)	
1	Good job
Expected Output	
1	Good job

11 . Question :

Function Description

Complete the findDay function in the editor below.

findDay has the following parameters:

- int: month
- int: day
- int: year

Returns

- string: the day of the week in capital letters

Input Format

A single line of input containing the space separated month, day and year, respectively, in *MM DD YYYY* format.

Constraints

- $2000 < year < 3000$

Code :

```
1 > import java.io.*; ...
12 import java.time.LocalDate;
13 class Result {
14
15     /*
16      * Complete the 'findDay' function below.
17      *
18      * The function is expected to return a STRING.
19      * The function accepts following parameters:
20      * 1. INTEGER month
21      * 2. INTEGER day
22      * 3. INTEGER year
23      */
24
25     public static String findDay(int month, int day, int year) {
26         java.time.LocalDate date= java.time.LocalDate.of(year , month, day);
27         return date.getDayOfWeek().name();
28     }
29 }
```

Output :

Congratulations!
You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Input (stdin)	
1	08 05 2015
Your Output (stdout)	
1	WEDNESDAY
Expected Output	
1	WEDNESDAY

12 . Question : Input Format

A single double-precision number denoting *payment*.

Constraints

- $0 \leq \text{payment} \leq 10^9$

Output Format

On the first line, print US: *u* where *u* is *payment* formatted for US currency.

On the second line, print India: *i* where *i* is *payment* formatted for Indian currency.

On the third line, print China: *c* where *c* is *payment* formatted for Chinese currency.

On the fourth line, print France: *f*, where *f* is *payment* formatted for French currency.

Code :

```
4  import java.math.*;
5  import java.util.regex.*;
6
7  public class Solution {
8
9      public static void main(String[] args) {
10         Scanner scanner = new Scanner(System.in);
11         double payment = scanner.nextDouble();
12         scanner.close();
13
14         Locale usLocale = Locale.US;
15         Locale indiaLocale = new Locale("en", "IN");
16         Locale chinaLocale = Locale.CHINA;
17         Locale franceLocale = Locale.FRANCE;
18
19         NumberFormat us=NumberFormat.getCurrencyInstance(usLocale);
20         NumberFormat india=NumberFormat.getCurrencyInstance(indiaLocale);
21         NumberFormat china=NumberFormat.getCurrencyInstance(chinaLocale);
22         NumberFormat france=NumberFormat.getCurrencyInstance(franceLocale);
23
24         System.out.println("US: " + us.format(payment));
25         System.out.println("India: " + india.format(payment));
26         System.out.println("China: " + china.format(payment));
27         System.out.println("France: " + france.format(payment));
28     }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

1 12324.134

Your Output (stdout)

```
1 US: $12,324.13
2 India: Rs.12,324.13
3 China: ¥12,324.13
4 France: 12 324,13 €
```

13 . Question :

The elements of a String are called characters. The number of characters in a String is called the length, and it can be retrieved with the String.length() method.

Given two strings of lowercase English letters, *A* and *B*, perform the following operations:

1. Sum the lengths of *A* and *B*.
2. Determine if *A* is lexicographically larger than *B* (i.e.: does *B* come before *A* in the dictionary?).
3. Capitalize the first letter in *A* and *B* and print them on a single line, separated by a space.

Input Format

The first line contains a string *A*. The second line contains another string *B*. The strings are comprised of only lowercase English letters.

Output Format

There are three lines of output:

For the first line, sum the lengths of *A* and *B*.

For the second line, write Yes if *A* is lexicographically greater than *B* otherwise print No instead.

For the third line, capitalize the first letter in both *A* and *B* and print them on a single line, separated by a space.

Code :

```
4 public class Solution {
5
6     public static void main(String[] args) {
7
8         Scanner sc=new Scanner(System.in);
9         String A=sc.next();
10        String B=sc.next();
11        int l1=A.length();
12        int l2=B.length();
13        System.out.println(l1+l2);
14        if (A.compareTo(B)>0){
15            System.out.println("Yes");
16        }else{
17            System.out.println("No");
18        }
19        String c1=A.substring(0,1).toUpperCase()+A.substring(1);
20        String c2=B.substring(0,1).toUpperCase()+B.substring(1);
21        System.out.println(c1+" "+c2);
22
23        sc.close();
24    }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 hello
2 java
```

Your Output (stdout)

```
1 9
2 No
3 Hello Java
```

14 . Question :

Given a string, s , and two indices, $start$ and end , print a [substring](#) consisting of all characters in the inclusive range from $start$ to $end - 1$. You'll find the String class' [substring method](#) helpful in completing this challenge.

Input Format

The first line contains a single string denoting s .

The second line contains two space-separated integers denoting the respective values of $start$ and end .

Constraints

- $1 \leq |s| \leq 100$
- $0 \leq start < end \leq n$
- String s consists of English alphabetic letters (i.e., $[a - zA - Z]$) only.

Output Format

Print the substring in the inclusive range from $start$ to $end - 1$.

Code :

```
1 import java.io.*;
2 import java.util.*;
3 import java.text.*;
4 import java.math.*;
5 import java.util.regex.*;
6
7 public class Solution {
8
9     public static void main(String[] args) {
10         Scanner in = new Scanner(System.in);
11         String S = in.next();
12         int start = in.nextInt();
13         int end = in.nextInt();
14         String str=S.substring(start,end);
15         System.out.println(str);
16         in.close();
17     }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

```
1 Helloworld
2 3 7
```

Your Output (stdout)

```
1 lowo
```

Expected Output

```
1 lowo
```

15. Question :

Function Description

Complete the getSmallestAndLargest function in the editor below.

getSmallestAndLargest has the following parameters:

- string *s*: a string
- int *k*: the length of the substrings to find

Returns

- string: the string ' + "\n" + ' where and are the two substrings

Input Format

The first line contains a string denoting *s*.

The second line contains an integer denoting *k*.

Constraints

- $1 \leq |s| \leq 1000$
- *s* consists of English alphabetic letters only (i.e., [a-zA-Z]).

Code :

```
1 > import java.util.Scanner; ...
4
5     public static String getSmallestAndLargest(String s, int k) {
6         String smallest = s.substring(0,k);
7         String largest = s.substring(0,k);
8
9         for (int i=1; i<=s.length()-k;i++){
10             String cs=s.substring(i,i+k);
11             if (cs.compareTo(smallest)<0){
12                 smallest=cs;
13             }
14
15             if (cs.compareTo(largest)>0){
16                 largest=cs;
17             }
18         }
19         return smallest + "\n" + largest;
20     }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Input (stdin)

```
1 welcometojava
2 3
```

Your Output (stdout)

```
1 ava
2 wel
```

16. Question :

A palindrome is a word, phrase, number, or other sequence of characters which reads the same backward or forward.

Given a string A , print Yes if it is a palindrome, print No otherwise.

Constraints

- A will consist at most 50 lower case english letters.

Sample Input

```
madam
```

Code :

```
1 import java.io.*;
2 import java.util.*;
3
4 public class Solution {
5
6     public static void main(String[] args) {
7
8         Scanner sc=new Scanner(System.in);
9         String A=sc.next();
10        String rev="";
11        for (int i=A.length()-1;i>=0;i--){
12            rev+=A.charAt(i);
13        }
14        if (A.equals(rev)){
15            System.out.println("Yes");
16        }else{
17            System.out.println("No");
18        }
19    }
20 }
```


Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

✓ Sample Test case 1

1 | **madam**

Your Output (stdout)

1 | **Yes**

17 . Question :

Input Format

There are two lines of input. The first line contains B : the breadth of the parallelogram. The next line contains H : the height of the parallelogram.

Constraints

- $-100 \leq B \leq 100$
- $-100 \leq H \leq 100$

Output Format

If both values are greater than zero, then the main method must output the area of the parallelogram. Otherwise, print "java.lang.Exception: Breadth and height must be positive" without quotes.

Code :

```
1  import java.io.*;
2  import java.util.*;
3  import java.util.Scanner;
4
5  public class Solution {
6
7      public static void main(String[] args) {
8          Scanner sc=new Scanner(System.in);
9          int b=sc.nextInt();
10         int h=sc.nextInt();
11         if (b>0 && h>0){
12             int a=b*h;
13             System.out.println(a);
14         }else{
15             System.out.println("java.lang.Exception: Breadth and height must be positive");
16         }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

✓ Sample Test case 1

1 | **1**
2 | **3**

Your Output (stdout)

1 | **3**

18 . Question :

Complete the isAnagram function in the editor.

isAnagram has the following parameters:

- string a: the first string
- string b: the second string

Returns

- boolean: If *a* and *b* are case-insensitive anagrams, return true. Otherwise, return false.

Input Format

The first line contains a string *a*.

The second line contains a string *b*.

Code :

```
1 > import java.util.Scanner; ...
4
5     static boolean isAnagram(String a, String b) {
6         if (a==null || b==null || a.length()!=b.length()){
7             return false;
8         }
9         a=a.toLowerCase();
10        b=b.toLowerCase();
11        int [] f=new int[26];
12        for (int i=0;i<a.length();i++){
13            f[a.charAt(i)-'a']++;
14            f[b.charAt(i)-'a']--;
15        }
16
17        for (int count : f){
18            if (count != 0){
19                return false;
20            }
21        }
22        return true;
23    }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

✓ Sample Test case 2

Input (stdin)

```
1  anagram
2  margana
```

Your Output (stdout)

```
1  Anagrams
```

19 . Question :

Input Format

A single string, s .

Constraints

- $1 \leq \text{length of } s \leq 4 \cdot 10^5$
- s is composed of any of the following: English alphabetic letters, blank spaces, exclamation points (!), commas (,), question marks (?), periods (.), underscores (_), apostrophes ('), and at symbols (@).

Output Format

On the first line, print an integer, n , denoting the number of tokens in string s (they do not need to be unique). Next, print each of the n tokens on a new line in the same order as they appear in input string s .

Code :

```
1  import java.io.*;
2  import java.util.*;
3
4  public class Solution {
5
6      public static void main(String[] args) {
7          Scanner scan = new Scanner(System.in);
8          String s = scan.nextLine();
9          s=s.trim();
10         if (s.length()==0){
11             System.out.println(0);
12             return;
13         }
14
15         String [] ts = s.split("[^A-Za-z]+");
16         System.out.println(ts.length);
17         for (String t : ts){
18             System.out.println(t);
19         }
20
21         scan.close();
22     }
23 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

Sample Test case 1

1 He is a very very good boy, isn't he?

Your Output (stdout)

```
1 10
2 He
3 is
4 a
5 very
6 very
7 good
8 boy
9 isn
```

20. Question :

Input Format

The first line of input contains an integer N , denoting the number of test cases. The next N lines contain a string of any printable characters representing the pattern of a regex.

Output Format

For each test case, print **Valid** if the syntax of the given pattern is correct. Otherwise, print **Invalid**. Do not print the quotes.

Code :

```
1  import java.util.Scanner;
2  import java.util.regex.*;
3
4  public class Solution
5  {
6      public static void main(String[] args){
7          Scanner in = new Scanner(System.in);
8          int testCases = Integer.parseInt(in.nextLine());
9          while(testCases>0){
10             String pattern = in.nextLine();
11             try{
12                 Pattern.compile(pattern);
13                 System.out.println("Valid");
14             }catch (PatternSyntaxException e){
15                 System.out.println("Invalid");
16             }
17             testCases--;
18         }
19         in.close();
20     }
21 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1  3
2  ([A-Z])(.+)
3  [AZ[a-z](a-z)
4  batcatpat(nat
```

Your Output (stdout)

```
1  Valid
2  Invalid
3  Invalid
```

21 . Code :

```
1 > import java.util.regex.Matcher; ...
16 class MyRegex{
17     String n="(\\d{1,2}|(0|1)\\d{2}|2[0-4]\\d|25[0-5])";
18     public String pattern = n+"\\. "+n+"\\. "+n+"\\. "+n;
19 }
20
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

```
1 000.12.12.034
2 121.234.12.12
3 23.45.12.56
4 00.12.123.123123.123
5 122.23
6 Hello.IP
```

Your Output (stdout)

```
1 true
2 true
3 true
4 false
```

22. Code :

```
5 public class DuplicateWords {
6
7     public static void main(String[] args) {
8
9         String regex = "\\b(\\w+)(?:\\s+\\1)+\\b";
10        Pattern p = Pattern.compile(regex, Pattern.CASE_INSENSITIVE);
11
12        Scanner in = new Scanner(System.in);
13        int numSentences = Integer.parseInt(in.nextLine());
14
15        while (numSentences-- > 0) {
16            String input = in.nextLine();
17
18            Matcher m = p.matcher(input);
19
20            // Check for subsequences of input that match the compiled pattern
21            while (m.find()) {
22                input = input.replaceAll(m.group(),m.group(1));
23            }
24
25            // Prints the modified sentence.
26            System.out.println(input);
27        }
28    }
29 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 5
2 Goodbye bye bye world world world
3 Sam went went to to to his business
4 Reya is is the the best player in eye eye game
5 in inthe
6 Hello hello Ab aB
```

Your Output (stdout)

```
1 Goodbye bye world
2 Sam went to his business
3 Reya is the best player in eye game
4 in inthe
```

23 . Code :

```
1 import java.util.Scanner;
2 class UsernameValidator {
3     /*
4      * Write regular expression here.
5      */
6     public static final String regularExpression = "[a-zA-Z][a-zA-Z0-9_]{7,29}$";
7 }
8
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 8
2 Julia
3 Samantha
4 Samantha_21
5 1Samantha
6 Samantha?10_2A
7 JuliaZ007
8 Julia@007
9 _Julia007
```

Your Output (stdout)

```
1 Invalid
```

24 . Code :

```
7  public class Solution{
8      public static void main(String[] args){
9
10         Scanner in = new Scanner(System.in);
11         int testCases = Integer.parseInt(in.nextLine());
12         String regex = "<([>]+)>([<]+)</\\1>";
13         Pattern pattern = Pattern.compile(regex);
14         while(testCases>0){
15             String line = in.nextLine();
16
17             Matcher matcher=pattern.matcher(line);
18             boolean found=false;
19             while (matcher.find()){
20                 System.out.println(matcher.group(2));
21                 found=true;
22             }
23             if (!found){
24                 System.out.println("None");
25             }
26
27             testCases--;
28         }
29     }
30 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

Download

✓ Sample Test case 1

✓ Sample Test case 2

```
1  4
2  <h1>Nayeem loves counseling</h1>
3  <h1><h1>Sanjay has no watch</h1></h1><par>So wait for a while<par>
4  <Ameesafat codes like a ninja</amee>
5  <SA premium>Imtiaz has a secret crash</SA premium>
```

Your Output (stdout)

```
1  Nayeem loves counseling
2  Sanjay has no watch
3  None
4  Imtiaz has a secret crash
```

25 . Code :

```
1  import java.io.*;
2  import java.math.*;
3  import java.security.*;
4  import java.text.*;
5  import java.util.*;
6  import java.util.concurrent.*;
7  import java.util.regex.*;
8
9
10
11  public class Solution {
12      public static void main(String[] args) throws IOException {
13          BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
14
15          String n = bufferedReader.readLine().trim();
16          BigInteger number = new BigInteger(n);
17          if (number.isProbablePrime(1)){
18              System.out.println("prime");
19          }else{
20              System.out.println("not prime");
21          }
22
23          bufferedReader.close();
24      }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

1 **13**

Your Output (stdout)

1 **prime**

Expected Output

1 **prime**

26. Code :

```
1 import java.util.*;
2 public class test {
3     public static void main(String[] args) {
4         Scanner in = new Scanner(System.in);
5         Deque deque = new ArrayDeque<>();
6         Set set = new HashSet<>();
7         int n = in.nextInt();
8         int m = in.nextInt();
9         int maxUnique = 0;
10
11         for (int i = 0; i < n; i++) {
12             int num = in.nextInt();
13             deque.add(num);
14             set.add(num);
15
16             if (deque.size() == m) {
17                 if (set.size() > maxUnique) {
18                     maxUnique = set.size();
19                 }
20                 if (maxUnique == m) {
21                     break;
22                 }
23                 int removed = (int) deque.removeFirst();
24                 if (!deque.contains(removed)) {
25                     set.remove(removed);
26                 }
27             }
28         }
29     }
30 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 6 3
2 5 3 5 2 3 2
```

Your Output (stdout)

```
1 3
```

Expected Output

```
1 3
```

27. Code :

```
7  ✓ public class Solution {
9  ✓      public static void main(String[] args) {
17 ✓          for (int i=0; i<m; i++){
18              String op = sc.next();
19              int p1 = sc.nextInt();
20              int p2 = sc.nextInt();
21
22 ✓          switch (op){
23 ✓              case "AND":
24                  bitsets[p1].and(bitsets[p2]);
25                  break;
26 ✓              case "OR":
27                  bitsets[p1].or(bitsets[p2]);
28                  break;
29 ✓              case "XOR":
30                  bitsets[p1].xor(bitsets[p2]);
31                  break;
32 ✓              case "FLIP":
33                  bitsets[p1].flip(p2);
34                  break;
35 ✓              case "SET":
36                  bitsets[p1].set(p2);
37                  break;
38              }
39              System.out.println(bitsets[1].cardinality()+" "+bitsets[2].cardinality());
40          }
41          sc.close();
}
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1  5 4
2  AND 1 2
3  SET 1 4
4  FLIP 2 2
5  OR 2 1
```

Your Output (stdout)

```
1  0 0
2  1 0
3  1 1
4  1 2
```

28 . Code :

```
5    import java.math.BigDecimal;...
14
15    // Sort array 's' of size 'n' in descending order
16    Arrays.sort(s, 0, n, new Comparator<String>() {
17        @Override
18        public int compare(String s1, String s2) {
19            BigDecimal b1 = new BigDecimal(s1);
20            BigDecimal b2 = new BigDecimal(s2);
21
22            // Compare b2 to b1 for descending order
23            return b2.compareTo(b1);
24        }
25    });
26 > //Output...
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

✔ Sample Test case 1

Input (stdin)

1	9
2	-100
3	50
4	0
5	56.6
6	90
7	0.12
8	.12
9	02.34
10	000.000

29 . Code :

```
1 import java.util.*;
2 public class Main{
3
4     static Iterator func(ArrayList mylist){
5         Iterator it=mylist.iterator();
6         while(it.hasNext()){
7             Object element = it.next();
8             if(element instanceof String)
9
10                break;
11        }
12        return it;
13
14    }
15    @SuppressWarnings({ "unchecked" })
16    public static void main(String []args){
17        ArrayList mylist = new ArrayList();
18        Scanner sc = new Scanner(System.in);
19        int n = sc.nextInt();
20        int m = sc.nextInt();
21        for(int i = 0;i<n;i++){
22            mylist.add(sc.nextInt());
23        }
24
25        mylist.add("###");
26        for(int i=0;i<m;i++){
27            mylist.add(sc.next());
28        }
29    }
30 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 2 2
2 42
3 10
4 hello
5 java
```

Your Output (stdout)

```
1 hello
2 java
```

30. Code :

```
10
11 ✓ public class Solution {
12 ✓     public static void main(String[] args) throws IOException {
13         BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
14
15         int[][] arr = new int[6][6];
16
17         // Read the 6x6 2D array
18 ✓     for (int i = 0; i < 6; i++) {
19         String[] rowItems = bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");
20 ✓         for (int j = 0; j < 6; j++) {
21             arr[i][j] = Integer.parseInt(rowItems[j]);
22         }
23     }
24
25     // Initialize max sum to minimum possible integer value
26     int maxHourglassSum = Integer.MIN_VALUE;
27
28     // Iterate through top-left corners of all possible 3x3 hourglasses
29 ✓     for (int i = 0; i <= 3; i++) {
30 ✓         for (int j = 0; j <= 3; j++) {
31             // Sum the 7 elements forming the hourglass
32 ✓             int currentSum = arr[i][j] + arr[i][j+1] + arr[i][j+2]
33                 + arr[i+1][j+1]
34                 + arr[i+2][j] + arr[i+2][j+1] + arr[i+2][j+2];
35         }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 1 1 0 0 0
2 0 1 0 0 0
3 1 1 1 0 0
4 0 9 2 -4 -4 0
5 0 0 0 -2 0 0
6 0 0 -1 -2 -4 0
```

Your Output (stdout)

```
1 13
```

Expected Output

```
1 13
```

31 . Code :

```
1  import java.io.*;
2  import java.util.*;
3  import java.text.*;
4  import java.math.*;
5  import java.util.regex.*;
6
7  public class Solution {
8
9      public static void main(String[] args){
10         Scanner scanner = new Scanner(System.in);
11
12         // Read size of array
13         int n = scanner.nextInt();
14         int[] arr = new int[n];
15
16         // Read array elements
17         for (int i = 0; i < n; i++) {
18             arr[i] = scanner.nextInt();
19         }
20
21         int negativeSubarrayCount = 0;
22
23         // Loop through every possible starting index 'i'
24         for (int i = 0; i < n; i++) {
25             int currentSum = 0;
26
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Input (stdin)

```
1  5
2  1 -2 4 -5 1
```

Your Output (stdout)

```
1  9
```

Expected Output

```
1  9
```

32 . Code :

```
1  import java.io.*;
2  import java.util.*;
3  import java.text.*;
4  import java.math.*;
5  import java.util.regex.*;
6
7  public class Solution {
8
9      public static void main(String[] args) {
10         Scanner scanner = new Scanner(System.in);
11
12         // Read the number of lines 'n'
13         int n = scanner.nextInt();
14
15         // 2D ArrayList to store rows of varying lengths
16         ArrayList<ArrayList<Integer>> lines = new ArrayList<>();
17
18         // Read each line
19         for (int i = 0; i < n; i++) {
20             int d = scanner.nextInt(); // Number of elements in current line
21             ArrayList<Integer> currentLine = new ArrayList<>();
22
23             for (int j = 0; j < d; j++) {
24                 currentLine.add(scanner.nextInt());
25             }
26
27             lines.add(currentLine);
28         }
29     }
30 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

Input (stdin)

```
1  5
2  5 41 77 74 22 44
3  1 12
4  4 37 34 36 52
5  0
6  3 20 22 33
7  5
8  1 3
9  3 4
10 3 1
11 4 3
12 5 5
```

33 . Code :

```
1  import java.util.*;
2
3  public class Solution {
4
5      public static boolean canWin(int leap, int[] game) {
6          return isSolvable(leap, game, 0);
7      }
8
9      private static boolean isSolvable(int leap, int[] game, int i) {
10         if (i >= game.length) {
11             return true;
12         }
13
14         if (i < 0 || game[i] == 1) {
15             return false;
16         }
17         game[i] = 1;
18         return isSolvable(leap, game, i + leap)
19             || isSolvable(leap, game, i + 1)
20             || isSolvable(leap, game, i - 1);
21     }
22
23     public static void main(String[] args) {
24         Scanner scan = new Scanner(System.in);
25         int q = scan.nextInt();
26         while (q-- > 0) {
27             int n = scan.nextInt();
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

```
1  4
2  5 3
3  0 0 0 0 0
4  6 5
5  0 0 0 1 1 1
6  6 3
7  0 0 1 1 1 0
8  3 1
9  0 1 0
```

Your Output (stdout)

```
1  YES
```


34 . Code :

```
2  import java.util.*;
3  import java.text.*;
4  import java.math.*;
5  import java.util.regex.*;
6
7  public class Solution {
8
9      public static void main(String[] args) {
10         Scanner scanner = new Scanner(System.in);
11
12         // Read initial list size N
13         int n = scanner.nextInt();
14
15         // Initialize dynamic List L
16         List<Integer> list = new ArrayList<>();
17         for (int i = 0; i < n; i++) {
18             list.add(scanner.nextInt());
19         }
20
21         // Read number of queries Q
22         int q = scanner.nextInt();
23
24         // Process each query
25         for (int i = 0; i < q; i++) {
26             String queryType = scanner.next();
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1  5
2  12 0 1 78 12
3  2
4  Insert
5  5 23
6  Delete
7  0
```

Your Output (stdout)

```
1  0 1 78 12 23
```

35 . Code :

```
1 //Complete this code or write your own from scratch
2 import java.util.*;
3 import java.io.*;
4
5 class Solution{
6     public static void main(String []argh)
7     {
8         Scanner in = new Scanner(System.in);
9
10        int n = in.nextInt();
11        in.nextLine();
12
13        Map<String, String> phoneBook = new HashMap<>();
14
15
16        for (int i = 0; i < n; i++) {
17            String name = in.nextLine();
18            String phone = in.nextLine();
19            phoneBook.put(name, phone);
20        }
21
22
23        while (in.hasNext()) {
24            String query = in.nextLine();
25
26            if (phoneBook.containsKey(query)) {
27                System.out.println(query + "=" + phoneBook.get(query));
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

```
8 uncle sam
9 uncle tom
10 harry
```

Your Output (stdout)

```
1 uncle sam=99912222
2 Not found
3 harry=12299933
```

Expected Output

```
1 uncle sam=99912222
2 Not found
3 harry=12299933
```

36 . Code :

```
2  class Solution{
4      public static void main(String []argh)
{
    8  while (sc.hasNext()) {
    9      String input=sc.next();
   10      Deque<Character> stack = new ArrayDeque<>();
   11      boolean isBalanced = true;
   12
   13      for (int i = 0; i < input.length(); i++) {
   14          char ch = input.charAt(i);
   15
   16          if (ch == '(' || ch == '{' || ch == '[') {
   17              stack.push(ch);
   18          } else {
   19              if (stack.isEmpty()) {
   20                  isBalanced = false;
   21                  break;
   22              }
   23
   24              char top = stack.pop();
   25              if ((ch == ')' && top != '(') ||
   26                  (ch == '}' && top != '{') ||
   27                  (ch == ']' && top != '[')) {
   28                  isBalanced = false;
   29                  break;
   30              }
   31      }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1  {}()
2  ({()})
3  {}(
4  []
```

Your Output (stdout)

```
1  true
2  true
3  false
4  true
```

37 . Code :

```
1  import java.io.*;
2  import java.util.*;
3  import java.text.*;
4  import java.math.*;
5  import java.util.regex.*;
6
7  public class Solution {
8
9      public static void main(String[] args) {
10         Scanner sc = new Scanner(System.in);
11
12         // Read input strings as BigInteger
13         BigInteger a = new BigInteger(sc.next());
14         BigInteger b = new BigInteger(sc.next());
15
16         // Compute sum and product
17         BigInteger sum = a.add(b);
18         BigInteger product = a.multiply(b);
19
20         // Output results
21         System.out.println(sum);
22         System.out.println(product);
23
24         sc.close();
25     }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

✓ Sample Test case 1

1	1234
2	20

Your Output (stdout)

1	1254
2	24680

38 . Code :

```
1  import java.util.*;
2
3  public class Solution {
4
5      public static void main(String[] args) {
6
7          Scanner scan = new Scanner(System.in);
8          int n = scan.nextInt();
9          int[] a = new int[n];
10         for (int i=0; i<n; i++){
11             a[i] = scan.nextInt();
12         }
13         scan.close();
14
15         // Prints each sequential element in array a
16         for (int i = 0; i < a.length; i++) {
17             System.out.println(a[i]);
18         }
19     }
20 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

1	5
2	10
3	20
4	30
5	40
6	50

Your Output (stdout)

1	10
2	20
3	30
4	40

39. Code :

```
1  ✓
2  import java.io.IOException;
3  import java.lang.reflect.Method;
4
5  class Printer
6  {
7      public <E> void printArray(E[] array){
8          for (E element : array){
9              System.out.println(element);
10         }
11     }
12 }
13
14
15  ✓ public class Solution {
16
17
18      public static void main( String args[] ) {
19          Printer myPrinter = new Printer();
20          Integer[] intArray = { 1, 2, 3 };
21          String[] stringArray = {"Hello", "World"};
22          myPrinter.printArray(intArray);
23          myPrinter.printArray(stringArray);
24          int count = 0;
25
26          for (Method method : Printer.class.getDeclaredMethods()) {
27              String name = method.getName();
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ **Sample Test case 0**

Your Output (stdout)

```
1  1
2  2
3  3
4  Hello
5  World
```

Expected Output

```
1  1
2  2
3  3
4  Hello
5  World
```

40 . Code :

```
1  import java.util.*;
2  class Checker implements Comparator<Player> {
3      @Override
4      public int compare(Player a, Player b) {
5          if (a.score == b.score) {
6              return a.name.compareTo(b.name);
7          }
8          return Integer.compare(b.score, a.score); // Descending order
9      }
10 }
11
12
13 > class Player{ ...
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1  5
2  amy 100
3  david 100
4  heraldo 50
5  aakansha 75
6  aleksa 150
```

Your Output (stdout)

```
1  aleksa 150
2  amy 100
3  david 100
4  aakansha 75
```

41 . Code:

```
25 public class Solution
27 public static void main(String[] args){
42
43 Collections.sort(studentList, new Comparator<Student>() {
44
45 public int compare(Student s1, Student s2) {
46
47 if (s1.getCgpa() != s2.getCgpa()) {
48 return Double.compare(s2.getCgpa(), s1.getCgpa());
49 }
50
51 if (!s1.getFname().equals(s2.getFname())) {
52 return s1.getFname().compareTo(s2.getFname());
53 }
54
55 return Integer.compare(s1.getId(), s2.getId());
56 }
57 });
58
59 for(Student st: studentList){
60 System.out.println(st.getFname());
61 }
62 }
63 }
```

Output :

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 5
2 33 Rumpa 3.68
3 85 Ashis 3.85
4 56 Samiha 3.75
5 19 Samara 3.75
6 22 Fahim 3.76
```

Your Output (stdout)

```
1 Ashis
2 Fahim
3 Samara
4 Samiha
```